Algebra 1 – Module 4 Test PRACTICE **WORKED KEY**

Module 4 — Linear Functions: Slope and Slope-Intercept Form

Answer each question, making sure to show your work in the space provided. Take the time to check your work. Write neatly. Circle your final answer.

1. Vocabulary. Write the letter of the term that matches each description. One term is never used. (4)

A) x-intercept B) slope C) y-intercept D) rise E) slope-intercept form

a. _____ the vertical change between two points

b. _____ the point where a line crosses the y-axis

c. _____ the ratio of rise to run; how steep a line is

d. _____ an equation written as y = mx + b

→ **a. D b. C c. B d. E**

2. Find the slope of the line through (2, −2) and (4, 0). (2)

rise +2, run +2

→ **m = 1**

3. Find the slope of the line through (0, 6) and (−8, 0). Simplify. (2)

rise −6, run −8

→ **m = 3/4**

4. Find each slope. (2)

(a) (−3, −1) and (−1, −1)

(b) (5, 6) and (5, −6)

→ **(a) 0 (rise is 0: horizontal) (b) undefined (run is 0: vertical)**

5. Name the slope m and the y-intercept b of each line. (2)

(a) y = −3 − 2x (b) y = 8

→ **(a) m = −2, b = −3 (b) m = 0, b = 8**

6. Write in slope-intercept form. (2)

2x + y = 3

subtract 2x on both sides

→ **y = −2x + 3**

7. Write in slope-intercept form. (2)

5x − 4y = 8

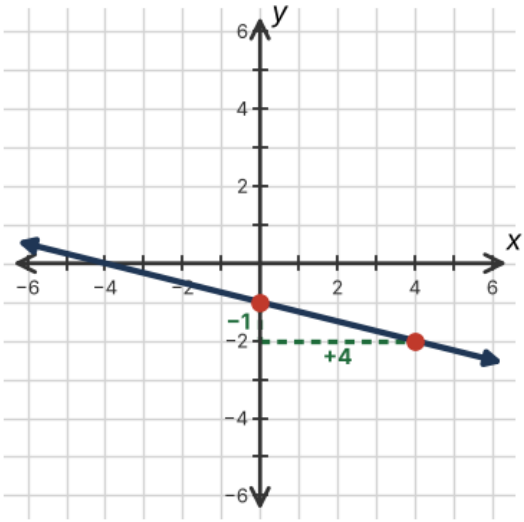
−4y = −5x + 8

divide every term by −4

→ **y = (5/4)x − 2**

8. Graph the line. (2)

y = −(1/4)x − 1

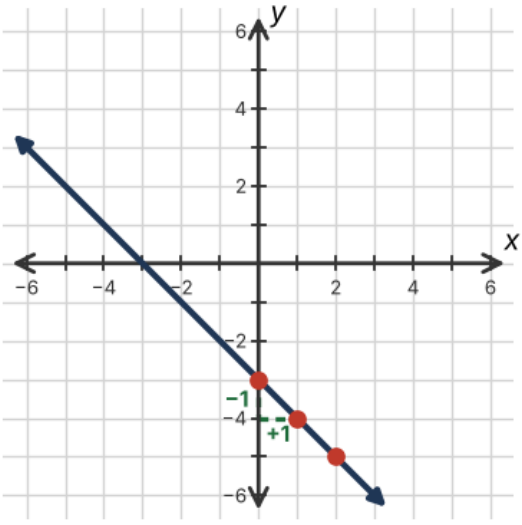


→ **b = −1, m = −1/4**

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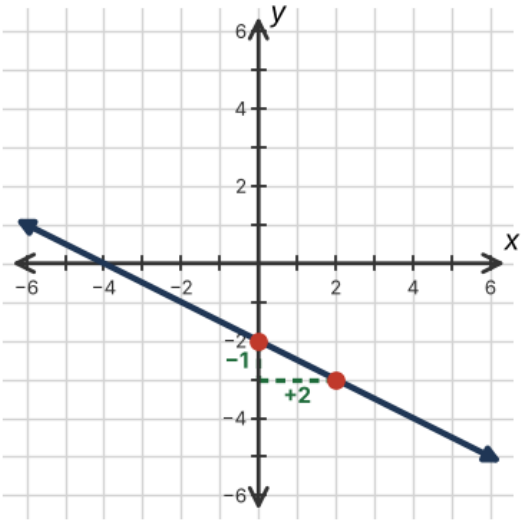
9. Write in slope-intercept form, then graph. (2)

x + y = −3



→ **y = −x − 3; b = −3, m = −1**

10. Write the equation of the line in slope-intercept form. (2)



b = −2; rise/run = −1/2

→ **y = −(1/2)x − 2**

11. A pool drains at a steady rate. After 2 hours it holds 520 gallons. After 5 hours it holds 400 gallons. Find the slope and the y-intercept, tell what each means, and write the equation (x = hours, y = gallons). (2)

slope = (400 − 520) / (5 − 2) = −40

b: 520 = −40(2) + b

→ **m = −40: it loses 40 gallons each hour; b = 600: gallons at the start; y = −40x + 600**

12. Error analysis. Marco rewrote 3x − 4y = 8 as y = (3/4)x + 2. Name the mistake in one sentence, then write the correct equation. (2)

→ **The constant has the wrong sign: dividing by −4 gives 8 ÷ (−4) = −2. Correct: y = (3/4)x − 2**

13. A line passes through (0, 6) and (2, 2). Write its equation in slope-intercept form. (2)

slope = −4/2 = −2; (0, 6) is the y-intercept

→ **y = −2x + 6**

Cumulative Review — ACT Style. Circle the best answer. No work required, but scratch work never hurts.

14. Solve: 3x − 3 = x + 3 (1)

(A) 0 (B) −3 (C) 5 (D) 3

→ **D (x = 3)**

15. f(x) = −4x − 6. Find f(−4). (1)

(F) −22 (G) 2 (H) 10 (J) −14

→ **H (10)**